

SYSTEMATIC REVIEWS

Efficacy of prescribed opioids for acute pain after being discharged from the emergency department: A systematic review and meta-analysis

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Abstract

Background: Opioids are often prescribed for acute pain to patients discharged from the emergency department (ED), but there is a paucity of data on their short-term use. The purpose of this study was to synthesize the evidence regarding the efficacy of prescribed opioids compared to nonopioid analgesics for acute pain relief in ED-discharged patients.

Methods: MEDLINE, EMBASE, CINAHL, PsycINFO, CENTRAL, and gray literature databases were searched from inception to January 2023. Two independent reviewers selected randomized controlled trials investigating the efficacy of prescribed opioids for ED-discharged patients, extracted data, and assessed risk of bias. Authors were contacted for missing data and to identify additional studies. The primary outcome was the difference in pain intensity scores or pain relief. All meta-analyses used a random-effect model and a sensitivity analysis compared patients treated with co-deine versus those treated with other opioids.

Trial registration Registered with Prospero #CRD42022332009.

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Results: From 5419 initially screened citations, 46 full texts were evaluated and six studies enrolling 1161 patients were included. Risk of bias was low for five studies. There was no statistically significant difference in pain intensity scores or pain relief between opioids versus nonopioid analgesics (standardized mean difference [SMD] 0.12; 95% confidence interval [CI] -0.10 to 0.34). Contrary to children, adult patients treated with opioid had better pain relief (SMD 0.28, 95% CI 0.13–0.42) compared to nonopioids. In another sensitivity analysis excluding studies using codeine, opioids were more effective than nonopioids (SMD 0.30, 95% CI 0.15–0.45). However, there were more adverse events associated with opioids (odds ratio 2.64, 95% CI 2.04–3.42).

Conclusions: For ED-discharged patients with acute musculoskeletal pain, opioids do not seem to be more effective than nonopioid analgesics. However, this absence of efficacy seems to be driven by codeine, as opioids other than codeine are more effective than nonopioids (mostly NSAIDs). Further prospective studies on the efficacy of short-term opioid use after ED discharge (excluding codeine), measuring patient-centered outcomes, adverse events, and potential misuse, are needed.

KEYWORDS

emergency department, opioids, outpatients, pain

INTRODUCTION

In 2019–2020, 8.1% of all emergency department (ED) visits in the United States resulted in an opioid prescription at discharge.¹ Though a decrease in opioid prescribing has been observed in recent years,^{2–4} ED physicians still prescribed opioids to treat moderate to severe acute pain at home.^{5–8} These prescription opioids are associated with adverse events (e.g., respiratory depression, constipation, nausea/vomiting, dry mouth, dizziness, drowsiness, weakness, pruritus)^{9–11} and can lead to long-term use,^{12,13} opioid misuse, and substance use disorder¹⁴ or death.¹⁵ However, alternatives such as nonsteroidal anti-inflammatory drugs (NSAIDs) are also associated with adverse events (e.g., gastrointestinal bleeding, acute stroke, myocardial infarction, congestive heart failure, cardiovascular death, hypertension, acute renal failure), particularly in older patients.^{16,17}

The efficacy of opioids for outpatient acute pain management has been well studied in the postoperative context. A systematic review on postoperative pain showed no pain intensity differences within a 24-hour follow-up period between opioids and NSAIDs and also between opioids and acetaminophen.¹⁸ A systematic review comparing codeine to NSAIDs in postoperative pain showed better pain scores and fewer adverse events in adult outpatients treated with NSAIDs.¹⁹ In a recent postsurgical randomized controlled dental pain study, naproxen (NSAID) was significantly more effective at reducing pain intensity and relief over 12 h than a combination of hydrocodone and acetaminophen.²⁰ Furthermore, an overview of systematic reviews on the management of mainly postoperative acute dental pain showed that NSAIDs were the best option.²¹ Postoperative opioids, particularly codeine or in the context of dental pain, do not seem to be superior to nonopioid analgesics.

The efficacy of opioids administered during an ED visit has also been well studied. For acute musculoskeletal injuries seen in the ED, opioids do not seem to offer a better benefit–harm ratio in relieving pain than NSAIDs.^{22–24} Others have found that opioids are not superior to NSAIDs or acetaminophen at reducing acute pain in a prehospital setting.²⁵ However, another literature review on musculoskeletal pain in the ED reported that “No interventions were more effective at decreasing pain intensity than opioids at 60 min.”²⁶ Furthermore, a prospective cohort study of ED patients discharged with an opioid prescription showed that patients using opioids were more likely to report pain relief than those using NSAIDs and/or acetaminophen.²⁷ Nonetheless, the study was observational, not controlled or randomized. Even so most of these studies do not address pain management after ED discharge. Analgesics administered only in the ED could provide a different effect at home after the initial ED treatment of acute pain.²⁸ Given that opioids are prescribed to ED-discharged patients for acute pain and the conflicting evidence regarding their short-term use, we aimed to synthesize the evidence regarding the efficacy of prescribed opioids compared to nonopioid analgesics for acute pain relief in ED-discharged patients.

METHODS

Review design

This systematic review and meta-analysis was prospectively registered (PROSPERO CRD42022332009). Its results are presented as per the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) guidelines (Data S1).²⁹ We also followed

the synthesis without meta-analysis reporting guidelines to complement the PRISMA guidelines.³⁰

Eligibility criteria

We included randomized controlled trials (RCT) and cluster RCTs designed to evaluate the efficacy of opioids prescribed for relief of acute pain in ED-discharged patients compared to nonopioid analgesics. All types of pain complaints were considered.

We excluded studies exclusively on patients with substance use disorder or chronic pain, evaluating only opioids given during the ED stay, comparing different opioids head-to-head, not reporting pain intensities or pain relief (after contacting authors), or available only as abstracts. There were no exclusions based on language or date of publication.

Data sources and search strategy

In collaboration with an information specialist, we developed a search strategy based on four search themes: “opioid,” “pain,” “emergency,” and “effectiveness” (detailed search strategies are available in Supplement Methods S1). The following databases were searched from their inception to May 5, 2022: CENTRAL, MEDLINE, EMBASE, CINAHL, and PsycINFO. An updated search was performed at the end of the process during the third week of January 2023 to collect recently published papers. We also searched [ClinicalTrials.gov](https://www.clinicaltrials.gov) and the International Standard Randomized Controlled Trial Number registry for ongoing studies. Gray literature was searched using Google Scholar and the first 300 hits were screened to identify any relevant studies. References from studies meeting inclusion criteria were examined and their authors were contacted to identify additional relevant studies.

Study selection

Identified references from all databases were transferred to Covidence systematic review manager (Covidence systematic review software, Veritas Health Innovation, available at www.covidence.org). Eight reviewers (RD, JP, MM, DW, GF, AF, AO, and AC) participated in a two-stage selection process of eligible studies. After duplicate removal, each citation was screened by two independent reviewers using the title and abstract for potential relevance. The full text of the remaining citations was then independently evaluated by two reviewers against inclusion and exclusion criteria. During both stages, a third reviewer (RD or AC) resolved discrepancies.

Data extraction

Data were extracted independently by two reviewers (JP and MM) using a standardized electronic sheet, and the conflicts were

resolved by consensus. Study authors were contacted when outcomes were incomplete or when ED patients were mixed in with other populations. Sample size, age range, inclusion/exclusion criteria, type of opioid, type of nonopioid analgesic, outcomes, and primary and secondary results were extracted.

Primary and secondary outcomes

Our primary outcome was the difference in pain intensity scores or pain relief between prescribed opioids and nonopioid analgesics at any time after an ED visit for acute pain. Our secondary outcomes were the percentage of patients with adverse events and the percentage of patients requiring rescue analgesics.

Risk-of-bias assessment

Risk of bias was assessed using the Effective Practice and Organization of Care (EPOC) risk-of-bias tool.³¹ Two reviewers (JP and MM) assessed the risk of bias independently, with conflicts resolved through discussion or involvement of a third reviewer (RD). RCTs were ranked as having low risk, being unclear, or having high risk of bias across nine domains (i.e., random sequence generation, allocation concealment, similarity of baseline outcome measurements, similarity of baseline characteristics, incomplete outcome data, knowledge of the allocated interventions adequately prevented during the study, protection against contamination, selective outcome reporting, and other risk of bias) and the overall risk of bias was derived. Risk-of-bias judgments for each domain across studies were summarized using risk-of-bias graphs and summaries. Unpublished studies were considered at high risk of bias.

Data synthesis and analysis

The difference in pain intensity scores (or in pain relief) between patients who were administered opioids versus those not administered opioids from baseline to cutoff time as specified in the original RCT (the final single time point from each follow-up were used for the main analysis), adverse events incidence, utilization of rescue analgesics, and data related to patient demographics were reported for each study.

If data were available for at least three studies, results for the main outcome and for adverse events were pooled and meta-analyzed. For the main efficacy outcome, studies that reported mean and standard deviation (SD) of pain relief for both opioid and nonopioid groups were expressed by standard mean differences (SMD) in forest plots. For studies that presented the percentage of patients with pain relief or with treatment failure, data were converted to SMD using a meta-analysis software (Comprehensive Meta-analysis: A Computer Program for Research Synthesis, Biostat, 2005).³² Data extracted for adverse events were all reported with the percentage of patients with

any side effects and were expressed in odds ratios (ORs). Sensitivity analyses were performed for studies with available short-term follow-up (≤ 3 days) to examine whether opioids were more effective in this situation. Another sensitivity analysis compared patients treated with codeine versus those treated with other opioids since codeine is a prodrug and its analgesic properties depend on its conversion to morphine, which varies considerably between subjects.³³ Sensitivity analyses were also performed excluding studies with high risk of bias and for studies involving a pediatric population compared to adults. Finally, we repeated the same analyses while excluding studies for which the comparator was only a placebo.

Heterogeneity was statistically assessed both overall and for each sensitivity analysis using I^2 , tau-square, and Cochran's Q tests. Prediction intervals were not presented since they can be problematic when the number of studies is small ($n < 10$).³⁴ A chi-square test was used to determine whether there was a difference between sensitivity categories. All effect sizes are reported with 95% confidence intervals (CIs). All analyses were conducted using an inverse variance weighting method and a random-effect model, even if the I^2 was low, due to the clinical heterogeneity of painful conditions, outcomes, and treatments.³⁵ Forest plots were performed using Revman, version 5.4 (The Nordic Cochrane Centre, The Cochrane Collaboration, 2014). No tests for publication bias were done (funnel plot asymmetry) because the power of the test is too low to distinguish chance from real asymmetry when there are fewer than 10 studies.³⁴

RESULTS

Search results and study characteristics

The initial search strategy generated 5942 references after duplicate removal. Of these, 47 were kept for full-text review and seven studies were retained. One of them³⁶ was further removed since it used dextropropoxyphene, an opioid that has been withdrawn from the market in 2009 because of its association with lethal events. Therefore, six studies^{37–42} were finally included in this systematic review and meta-analysis (Figure 1). One study³⁹ compared two types of opioids (oxycodone and tramadol) to a placebo. Patients could consume rescue medications and placebo has demonstrated analgesic effects^{43,44}; we kept only the oxycodone versus placebo comparison as tramadol has nonopioid analgesic effects.⁴⁵ Another study⁴² compared oxycodone to two nonopioids (NSAID and cyclobenzaprine); we kept only the oxycodone versus NSAIDs comparison since cyclobenzaprine is mainly a muscle relaxant.⁴⁶ One conflict on data extraction for the outcome measured in the study by Drendel et al.⁴⁰ was settled through consensus. Four differences in risk of bias assessment were observed between the two raters, two in the protection against contamination and two in the selective outcome reporting domain. These were resolved through consensus by the two raters.

Four included studies^{37,39,40,42} were conducted in the United States and two^{38,41} in Canada; four studies^{37–39,42} included adult

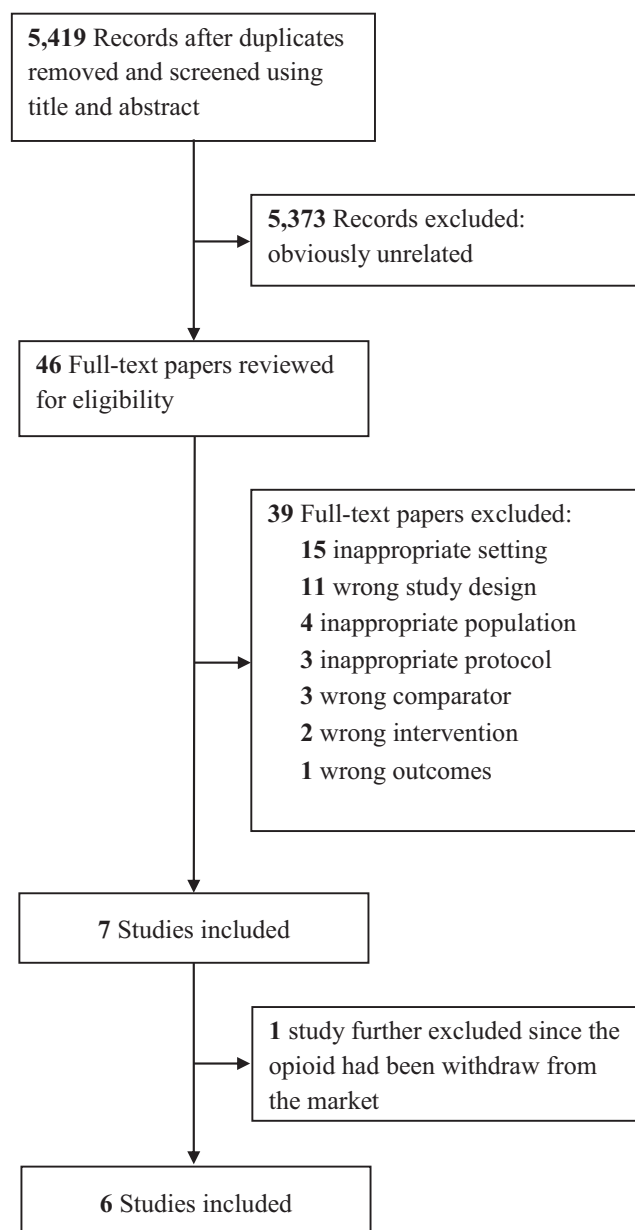


FIGURE 1 PRISMA flowchart of included studies.

and two others^{40,41} included pediatric populations (Table 1). All studies included patients with musculoskeletal pain; five of them^{37,38,40–42} had NSAIDs as a comparator and one had a placebo.³⁹ Three studies^{37,38,40} used codeine as the opioid, and the three others used hydrocodone,³⁹ morphine,⁴¹ or oxycodone.⁴² Except for morphine, opioids were combined with acetaminophen and their dosage and administration frequency were not equivalent. The type of NSAIDs used also varied across studies: diflunisal, ketorolac, ibuprofen (two studies), and naproxen. Follow-up duration varied from 1 to 7 days and outcomes were pain relief for five studies^{37–39,41,42} and treatment failure (use of rescue medication) for the other.⁴⁰ The majority of studies applied extensive exclusion criteria (Supplement Table S2).

TABLE 1 Characteristics of the studies included.

First author, year, country	Sample size, M/W or B/G	Age range (years)	Inclusion criteria	Opioid treatment (No. assigned)	Nonopioid treatment (No. assigned)	Measured outcomes	Follow-up duration (days)
Aghababian, ³⁷ 1998, United States	N=40, 24M/16W	18–51	Grade 2 ankle sprain with mild or moderate pain	Codeine [30mg] with acet [300mg], one or two tablets every 4 h as needed (n=21)	Diflunisal [1000 mg], followed by 500mg every 8–12 h as needed (n=19)	Self-evaluation form, record every 12 h degree of pain, tenderness, swelling, and limitation in motion, on a scale of 0 = none, 1 = mild, 2 = moderate, 3 = severe; to inform investigator of adverse events or adverse reactions to medication	7
Innes, ³⁸ 1998, Canada	N=122, 96M/26W	19–62	Adults with moderate or severe musculoskeletal low-back pain of less than 72 h duration, well enough to be discharged within 2–4 h	Codeine [60mg] with acet [600mg], one tablet every 4–6 h as needed, up to six doses per 24 h (n=60)	Ketorolac [10mg], one tablet every 4–6 h as needed, up to four doses per 24 h; those requiring a fifth or sixth analgesic dose in any 24 h period were given acet, 650 mg per dose, in each of two optional doses (n=63)	Pain intensity visual analog scale 0–100; pain intensity category: none, mild, moderate, severe, very severe; pain relief: none, a little, some, a lot, complete; adverse events	7
Hewitt, ³⁹ 2007, United States	N=596, 315M/281W	18–81	Adults with an ankle sprain within the previous 48 h, pain score ≥ 50 on a visual analog scale of 0–100, and pain intensity of moderate to severe on a four-category scale	Hydrocodone [7.5 mg] with acet [650 mg], one or two tablets up to four times a day as needed (n=201) Tramadol [37.5 mg] with acet [325 mg], one or two tablets up to four times a day as needed (n=190)	Placebo; one or two tablets up to four times a day as needed (n=205)	100-mm pain visual analog scale; the pain intensity numeric rating scale from 0 to 3; pain relief numeric rating scale of –1 (worse), 0 (none), 1 (slight relief), 2 (moderate relief), 3 (a lot of relief), 4 (complete relief), adverse events	5
Drendel, ⁴⁰ 2009, United States	N=244, 126B/118G	4–18	Acute pediatric arm fracture, without reduction or manipulation, not an open fracture	Acet with codeine [120mg/5mg per 5mL] was dosed according to the codeine content of 1 mg/kg every 4–6 h as needed (n=116)	Ibuprofen suspension [100 mg/5 mL] was dosed at 10 mg/kg every 4–6 h as needed (n=128)	Use of rescue medication for inadequate pain relief, daily adverse effects	3
Poonai, ⁴¹ 2014, Canada	N=134, 42B/92G	5–17	Children with nonoperative, radiographically evident extremity fracture sustained within the preceding 24 h	Morphine [0.5 mg/kg, max 10mg], one tablet every 6 h as needed, up to four doses per 24 h (n=66)	Ibuprofen [10 mg/kg, max 600 mg], one tablet every 6 h as needed, up to four doses per 24 h (n=68)	Differences in faces pain scale from immediately before and 30 min after each dose, from 0 (no pain) to 5 (maximum amount of pain); adverse effects; no of acet rescue dose	1
Friedman, ⁴² 2015, United States	N=323, 165M/158W	21–64	Adult acute lower back pain nontraumatic or nonradicular for less than 2-week duration with a score ≥ 5 on the Roland-Morris Disability Questionnaire	Oxycodone [5 mg] with acet [325 mg], one or two tablets every 8 h as needed, naproxen [500mg] twice a day (n=108)	Cyclobenzaprine [5mg], one or two tablets every 8 h as needed, naproxen [500mg] twice a day (n=108) placebo; one or two tablets every 8 h as needed, naproxen [500mg] twice a day (n=107)	Worst low-back pain in the past 24 h (mild/none, moderate/severe), adverse effects	7

Abbreviations: Acet, acetaminophen; B, boys; G, girls; M, men; W, women.

Risk-of-bias assessment

One included RCT study³⁷ was at high overall risk of bias; the remaining five^{38–42} were at low risk of bias (EPOC; Supplement Figure S1).

Descriptive results

The main and secondary outcomes are reported in Table 2. Five^{37,38,40–42} of six studies did not report a significant difference in efficacy between opioids and NSAIDs, while Hewitt et al.³⁹ reported a greater average pain relief for the opioid group compared to placebo. All included studies reported a higher rate of adverse events for patients using opioids compared to those

using NSAIDs or placebo. Two studies reported information on rescue medication; one study³⁹ showed that more patients used rescue medication in the placebo group compared to those in the opioid group, and the other⁴¹ showed that patients in the opioid group used rescue analgesic more often than those assigned to the NSAID group.

Meta-analysis

The data for opioid efficacy and side effects analysis from the six studies were available for meta-analysis. The number of included patients was 584 for the opioid group and 577 for the nonopioid group. The forest plot showed that opioids were not more effective than

TABLE 2 Results of the studies included.

First author, year	Primary results	Adverse events	Rescue analgesics	Conclusions
Aghababian, ³⁷ 1986	Mean $\pm$ SD pain relief, Day 7: 1.16 $\pm$ 0.5 diflunisal 1.19 $\pm$ 0.6 codeine/acet	None of the diflunisal group reported adverse events compared to 29% of patients with codeine/acet group	Not reported	Diflunisal was as effective as codeine and acet; no statistical tests reported, descriptive results only
Innes, ³⁸ 1998	After Day 1, 24% of ketorolac and 31% of codeine/acet had a lot or complete pain relief After Day 4, 53% of ketorolac and 55% of codeine/acet had a lot or complete pain relief	34% in the ketorolac group reported at least one adverse event compared to 64% of the patients in the codeine/acet group ($p < 0.001$)	Not reported	No significant differences between groups for overall pain relief at 1 week, no significant differences between study groups in overall drug rating
Hewitt, ³⁹ 2007	Mean $\pm$ SD pain relief, Day 1: 5.4 $\pm$ 3.65 Placebo 6.8 $\pm$ 3.62 hydrocodone/acet Mean $\pm$ SD pain relief, Days 1–5: 1.7 $\pm$ 1.096 placebo 2.1 $\pm$ 1.085 hydrocodone/acet	19.3% in the placebo group reported at least one adverse events compared to 36.5% of patients in hydrocodone/acet group ($p < 0.001$)	Efficacy failure defined as need of supplemental analgesic or stopping actual medication was 11.2% for placebo, 2.5% for hydrocodone/acet	Groups with hydrocodone reported significant greater average pain relief than placebo
Drendel, ⁴⁰ 2009	Proportion of treatment failure for lack of pain relief: 20.3% Ibuprofen, 31.0% codeine/acet	29.5% of the patients reported any adverse effect in the ibuprofen group compared to 50.9% in the codeine/acet group (significant difference)	If pain relief was inadequate after 1 h, use the other medication (see primary results)	No significant differences between both groups on treatment failure
Poonai, ⁴¹ 2014	Mean $\pm$ SD pain relief day, first dose: 1.3 $\pm$ 1 ibuprofen 1.5 $\pm$ 1.2 morphine	30.9% in the ibuprofen group reported any adverse effect compared to 56.1% of the patients in the morphine group ($p < 0.01$)	26% of the patients had acet rescue dose for morphine group compared to 15% in the ibuprofen group ($p = 0.10$)	No significant difference in analgesic efficacy between morphine and ibuprofen
Friedman, ⁴² 2015	Worst low-back pain in the past 24 h of Day 7 (mild/none): 54% for naproxen group and 65% for oxycodone/acet/naproxen group	21% of the patients reported any adverse effect in the naproxen group compared to 40% in oxycodone/acet/naproxen group (significant difference)	Not reported	No significant differences between the oxycodone/acet/naproxen vs. naproxen groups

Abbreviation: Acet, acetaminophen.

nonopioids in the treatment of acute musculoskeletal pain after ED discharge (SMD 0.12, 95% CI -0.10 to 0.34, $I^2=62\%$; Figure 2). Furthermore, patients treated with opioids experienced a higher rate of side effects than those treated with NSAIDs or placebo (OR 2.64, 95% CI 2.04–3.42, $I^2=0\%$; Figure 3).

Sensitivity analyses

The results of opioid efficacy for studies with data on a short-term follow-up (≤ 3 days)^{38–41} or without the study at high risk of bias³⁷ were similar as those reported from the main analysis (SMD 0.12, 95% CI -0.20 to 0.45, $I^2=77\%$; and SMD 0.12, 95% CI -0.12 to 0.37, $I^2=69\%$; Supplement Figures S2 and S3). Adult patients treated with opioids had better pain relief than those treated with nonopioids (SMD 0.28, 95% CI 0.13–0.42, $I^2=0\%$; Supplement Figure S4) contrary to children who had nonsignificant difference between opioids and nonopioids ($p=0.78$). However, the comparison of studies with pediatric versus adult patients showed no statistically significant difference between these subgroups ($\chi^2=1.78$, $p=0.18$). When we removed the study³⁹ that had a placebo as a comparator in the adult subgroup, the effect was similar but was no longer statistically significant (SMD 0.16, 95% CI -0.06 to 0.38, $I^2=0\%$; Supplement Figure S5).

The comparison of studies using codeine versus opioids other than codeine showed a significant difference between these subgroups ($\chi^2=7.47$, $p=0.006$). Codeine was not more effective than nonopioid medications (SMD -0.12, 95% CI -0.38 to 0.14,

$I^2=19\%$). However, patients treated with other opioids than codeine had better pain relief than those treated with nonopioids (SMD 0.30, 95% CI 0.15–0.45, $I^2=0\%$; Figure 4). Results without the only study at high risk of bias³⁷ were identical for opioids other than codeine since that study was done with codeine (Supplement Figure S6). Also, when removing the study³⁹ that had a placebo as a comparator in the opioids other than codeine subgroup, the direction of the effect remained but was no longer statistically significant ($p=0.06$, SMD 0.22, 95% CI -0.01 to 0.44, $I^2=0\%$; Supplement Figure S7).

DISCUSSION

Main finding summary

This meta-analysis of studies originating from the United States and Canada on adult and pediatric populations showed that opioids are not more effective than nonopioids (mostly NSAIDs) in managing acute pain after ED discharge. However, a sensitivity analysis showed that opioids other than codeine provided better pain relief than nonopioids. Additionally, opioids were associated in all studies with a higher rate of adverse events and in one study with higher use of rescue medications.⁴¹ While we planned to include RCTs on all types of acute outpatient pain management after ED discharge, we found only studies including patients with musculoskeletal injuries. We did not observe a significant difference in results between adult and pediatric populations.

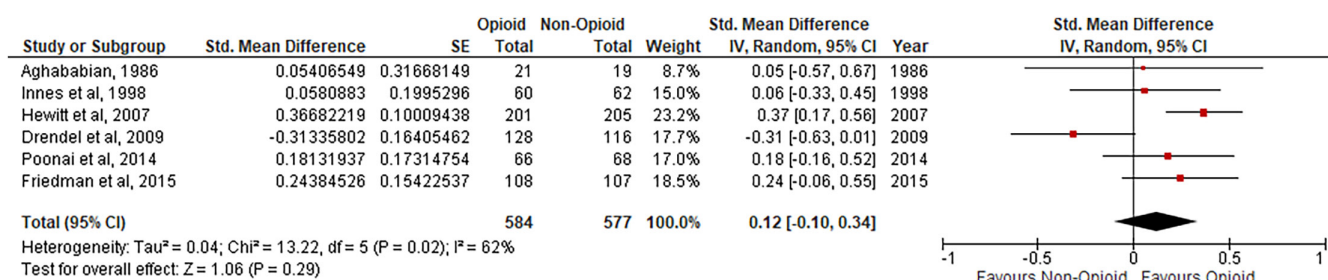


FIGURE 2 Forest plot of opioid efficacy after ED discharge.

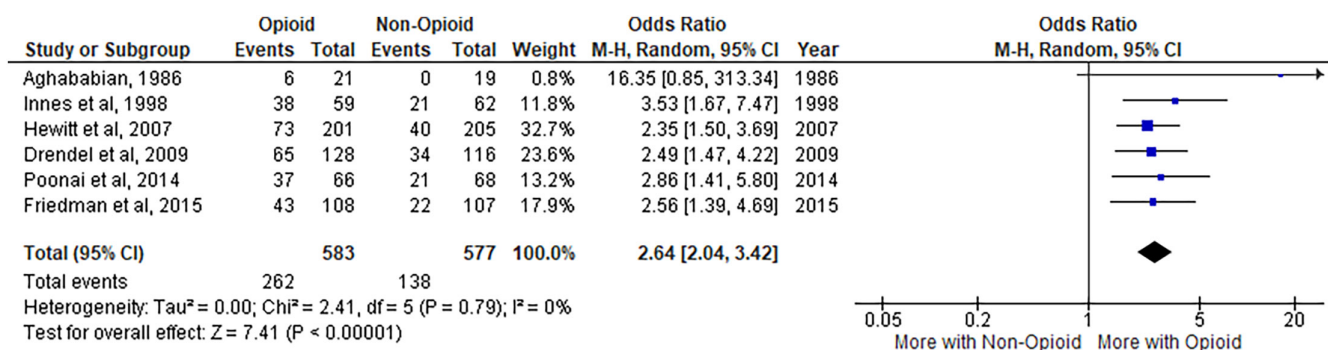


FIGURE 3 Forest plot of adverse events after ED discharge.

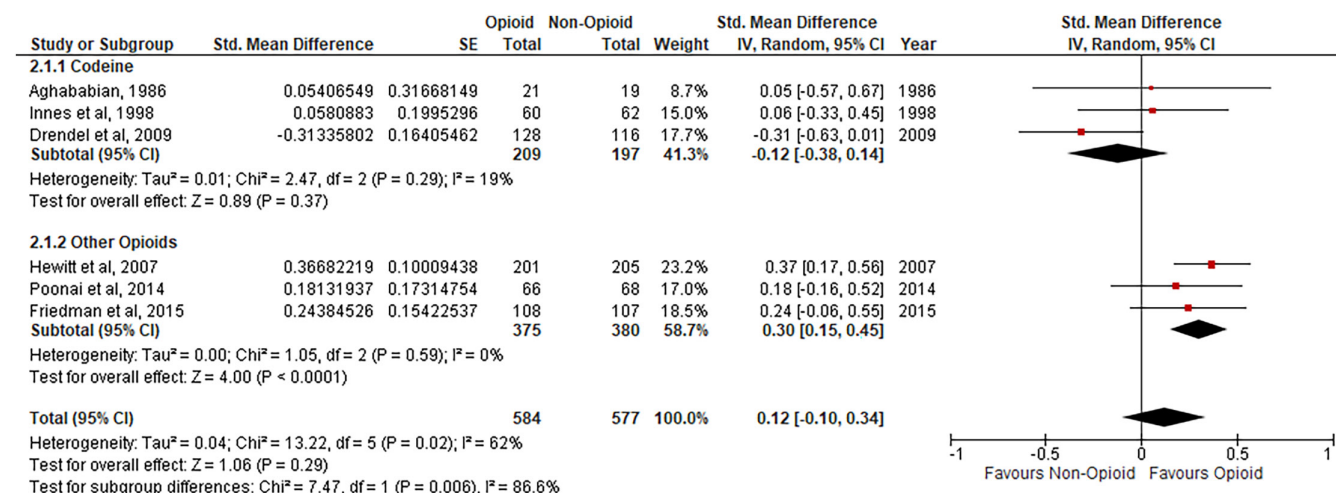


FIGURE 4 Forest plot of opioid efficacy for codeine versus other opioids.

Risk of bias

Five studies^{38–42} were at low risk of bias and only one study³⁷ on adults with an ankle sprain was at high risk of bias because of the lack of intervention allocation concealment and different baseline characteristics. Nevertheless, after removing this study in a sensitivity analysis, results were similar which demonstrates their consistency. There was significant heterogeneity in results from studies on opioid efficacy, but this was mostly due to the poor analgesic effect of codeine. Heterogeneity was very low for studies using opioids other than codeine, as it was for adverse events associated with opioids in general. Furthermore, one study⁴² on acute low-back pain found that adding oxycodone/acetaminophen to naproxen alone did not improve functional outcomes or pain at 1-week follow-up. In this study the oxycodone/acetaminophen was prescribed every 8 h as needed; since the elimination half-life of oxycodone is 3.51 h (SD ± 1.43)⁴⁷ and that oxycodone is typically given every 4 to 6 h,⁴⁸ this could explain the ineffectiveness of oxycodone.

Comparison with previous systematic reviews

A 2022 systematic review examined the impact of opioids on adults during and after their ED stay.⁴⁹ They concluded that, “Opioids may have pain outcomes equivalent to those of NSAIDs, but evidence on comparisons of harms is very uncertain and heterogeneous.” However, they did not perform a meta-analysis. Nevertheless, they only included two studies on opioids prescribed upon ED discharge, which were included in our study.^{37,42} We identified two other adult studies^{38,39} with data on opioids after discharge from the ED that they included only in the “opioids prescribed in the ED” section. A systematic review published in 2020 evaluated outpatient acute pain management (medication and other approaches) from non-low-back musculoskeletal injuries in adults and found that opioids combined with acetaminophen compared to placebo improved pain from Day 1 to 7, but not for immediate pain (less than 2 h), and caused more adverse events.²² However, they included all parallel-design studies on musculoskeletal injuries in

any outpatient setting and studies with a single dose of opioids; results could be different for acute pain management in an ED-discharged population. The efficacy of treatment (medication and other approaches) for acute and subacute nonspecific low-back pain has been examined in another systematic review.⁵⁰ They concluded that to reduce pain intensity, the most efficacious interventions (up to 1 week) were heat wrap, manual therapy, exercise, NSAIDs, and opioids; mild or moderate adverse events were most often associated with opioids, NSAIDs, and steroids. However, they identified only three studies on opioids; two were performed in the ED and one was conducted in an outpatient setting using meptazinol,⁵¹ which is seldomly used and available only in a few European countries.

Systematic reviews on specific painful conditions obtained results aligned with ours. For osteoarthritis pain, one review concluded with overall low confidence that the effect of NSAIDs for knee pain was similar to that of opioids.⁵² However, another systematic review on osteoarthritis not limited to the knee found that opioids may provide a small benefit in pain and disability but may also increase the risk of adverse events. As they did not encounter a study using codeine, their results are comparable to our sensitivity analysis excluding codeine.⁵³ Moreover, a systematic review on postoperative pain demonstrated that codeine provides less pain relief and more adverse events than NSAIDs.¹⁹ Another study, evaluating codeine/acetaminophen and oxycodone/acetaminophen combinations on the first day after ED discharge, demonstrated a not statistically significant trend in more pain relief with oxycodone/acetaminophen (66% vs. 61%).⁵⁴ However, one review examined the outpatient treatment of acute pain in general (surgical and nonsurgical) and concluded that opioid therapy was associated with similar efficacy as an NSAID and increased risk of short-term adverse events.¹⁸

Opioid prescribing guidelines

Many recent studies showed that NSAIDs or acetaminophen are as good as or better than opioids for the treatment of acute pain.^{18–25}

Due to the opioid crisis and the postulate that short-term opioids are ineffective and associated with a high risk of abuse, some health care professionals have stopped using opioids in the ED.⁵⁵ Most opioid prescribing guidelines for chronic noncancer pain do not adequately guide ED acute care.⁵⁶ The American College of Emergency Physicians 2020 clinical policy on adult opioid prescribing presented Level C recommendations to preferentially prescribe nonopioid analgesics for acute painful conditions of ED-discharged patients.⁵⁷ However, they also state that excessive limitation of opioid use in that context may lead to inadequate pain management. A position statement by the American Academy of Emergency Medicine suggests using a short-course (2–3 days) regimen of opioids, while excluding some medical conditions (e.g., chronic obstructive pulmonary disease, sleep apnea, substance use disorder) and being mindful of total acetaminophen exposures with combination opioids.⁵⁸ The Canadian Association of Emergency Medicine provides a guideline for ED management of opioid use disorder, but they do not provide guidance on ED opioid prescribing.⁵⁹ In our review, codeine did not provide better pain relief than nonopioids. However, codeine combined with acetaminophen is often selected has a low-potency opioid even if some clinicians propose to completely stop using it.^{60,61} Furthermore, a prospective cohort study of ED patients discharged with an opioid prescription, which excluded codeine or combination with acetaminophen, showed that patients using opioids were more likely to report pain relief than those using NSAIDs and/or acetaminophen, even if they reported more severe baseline pain intensity.²⁷ This approach allows optimal dosage of both acetaminophen and the opioid and in the same cohort less than 1% of patients misused opioids at 3 months.⁶² Yet, the risk of long-term use or misuse after a short course of opioids for acute pain remains poorly studied. Additionally, not using or using low-potency opioids for acute post-ED pain may influence the development of chronic pain, as some pain trajectories in the 2 weeks after an ED visit are associated with chronic pain.⁶³

There is a need to develop quality research and effective strategies to optimize, but not necessarily outright eliminate, opioid prescribing upon ED discharge. For example, it might be beneficial for selected patients to be prescribed a small quantity of a specific type of opioid (not combination with acetaminophen), which could be served in even smaller portions by the pharmacist and associated with optimal dosage of acetaminophen and/or NSAIDs.^{64,65} Such strategies will require prospective research to further delineate appropriate indications and optimize risk stratification, type of opioid, and prescription method/duration. These studies should measure patient-centered outcomes and ensure equitable treatment for all ED-discharged patients.

LIMITATIONS

Our review is limited by the study population's heterogeneity (pediatric and adult), medications type/dosage, and follow-up duration. However, analyses were done using a random-effect

model, which partially compensated for this diversity and after excluding codeine, the heterogeneity was very low ($I^2=0\%$). Furthermore, results were consistent in sensitivity analyses on the adult population and in which the study with high risk of bias and the one with only placebo as a comparison was excluded. Also, we studied only opioid use after ED discharge; results could be different in another context. Most studies utilized extensive exclusion criteria and included only patients with musculoskeletal injuries in a North American population, which limit their external validity. The adult studies included only painful syndromes of sprains and strains, and none had fractures that might have been better relieved by opioids. Most included studies (five of six) used a combination of acetaminophen (one also added an NSAID) and opioid (two of them at higher acetaminophen doses than what is currently permitted) instead of opioid alone, which may have contributed to opioid efficacy. Finally, included studies were not designed to assess the risk of overdose, opioid use disorder, or long-term harms.

CONCLUSIONS

In summary, this meta-analysis demonstrated that for ED-discharged patients with acute musculoskeletal pain, opioids do not seem more effective than nonopioid analgesics. However, this absence of efficacy seems to be driven by codeine, as other opioids are more effective than nonopioids (mostly nonsteroidal anti-inflammatory drugs) but also associated with more adverse events. Further prospective studies on the efficacy of short-term opioid use after ED discharge (excluding codeine), measuring patient-centered outcomes, adverse events, and potential misuse, are needed.

AUTHOR CONTRIBUTIONS

Raoul Daoust, Jean Paquet, and Alexis Cournoyer designed the study. Raoul Daoust, Jean Paquet, and Martin Marquis designed the search strategy. Raoul Daoust, Jean Paquet, Martin Marquis, Jean-Marc Chauny, David Williamson, and Alexis Cournoyer assessed study eligibility. Jean Paquet and Martin Marquis assessed the included studies' quality. Jean Paquet and Martin Marquis extracted the data. Jean Paquet performed the statistical analyses. Raoul Daoust wrote the first draft of the manuscript. All authors including Suneel Upadhye and Justine Lessard contributed to the interpretation and subsequent edits of the manuscript. Raoul Daoust takes responsibility for the paper.

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CONFLICT OF INTEREST STATEMENT

The authors declare no competing interests.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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